

11. FERTILISATION

DURATION : 14:26

STUDY SHEET

COURSE SUMMARY

This session deeply explores the fertilisation process, from ovulation to the reception of the sperm by the oocyte. The concepts of ovarian mobility, natural barriers to fertilisation, and membrane recognition through ZP3 are discussed to understand the complex dynamics of reproduction. The session also highlights the implications of in vitro fertilisation techniques, emphasising essential emotional and energetic links between mother and child, as well as the importance of knowing one's personal history for holistic health.

FERTILISATION

OVULATION AND OVARIAN MOBILITY

Ovulation is a crucial process, and it is essential to understand **ovarian mobility**. The fimbriae of the ovary play a key role in this mechanism.

The anatomy of the lesser pelvis, uterus, and ovaries is complex. Structures such as the **broad ligaments**, **round ligaments**, and **lumbo-ovarian ligaments** are found there, all composed of the same tissue.

The embryology and anatomy of the uterus reveal that it is a fold of the **posterior parietal peritoneum**. This structure must be mobile to capture the **oocyte**. A hemispherical movement, linked to motility, is necessary for this capture.

If we consider the digestive system, **Tol's fascia** inserts into the posterior parietal peritoneum. In cases of **appendicitis**, for example, a loss of energy or fascial disturbance can affect ovarian mobility and prevent oocyte capture.

The peritoneum must be free to allow fertilisation. If the oocyte is not captured, it can remain in the pelvic area, leading to dangerous **ectopic pregnancies**. There is a direct link between the vagina and the peritoneal cavity, which highlights the importance of this mobility.

NATURAL BARRIERS TO FERTILISATION

In males, fertilisation is complicated by the absence of connection between the reproductive structures, due to the **Müllerian tubercle** and the **Wolffian duct**.

In females, several natural barriers exist. The first is **vaginal acidity**, which helps destroy unwanted bacteria. Then, a **mucus plug** at the cervix plays a crucial role. This plug opens once a month, allowing sperm to pass. If it does not open, passage is blocked.

Once in the uterus, sperm must navigate through **cilia** that help them move forward. They are attracted by heat, sugar, and electrical signals. However, it is not a single sperm that succeeds, but a group that manages to reach the oocyte.

MEMBRANE RECOGNITION: ZP3

When the sperm reaches the oocyte, it must go to the most optimal area, where the **oocyte** is ready to receive the sperm. On the oocyte membrane are **glycoproteins** called **ZP3**, which act as a recognition key.

This membrane recognition is essential: if it fails, fertilisation cannot occur.

IN VITRO FERTILISATION AND ITS IMPLICATIONS

In vitro fertilisation (IVF) is a common method, particularly **ICSI** (Intracytoplasmic Sperm Injection), where sperm is directly injected into the oocyte. This technique bypasses natural recognition, which can lead to long-term fertility problems.

Children born from IVF may present complications. During consultations, it is important to ask questions about conception, especially about any miscarriages. Uteri can be charged with residual energy related to these events, requiring energetic cleansing.

It is crucial to recognise the child, establishing a bond between mother and child. Mothers must establish physical and emotional contact with their child to foster this connection.

THE IMPORTANCE OF KNOWING ONE'S HISTORY

Knowing one's history is fundamental. A child born from sperm donation, for example, may carry a lie within them, unaware of their origins. Tissues do not lie, and an osteopath can feel this truth through their hands.

THE ROLE OF OOCYTE POLARISATION

The **oocyte** is polarised. When the sperm arrives, it causes a reorganisation of the oocyte's axes, creating dorsal-ventral and apico-caudal axes. This symbolic moment is a connection with the primitive heart.

ATTACK ON THE OOCYTE MEMBRANE

Sperm must pass through several layers to reach the **zona pellucida**. Once on the membrane, membrane recognition must occur. The sperm that successfully binds releases an acrosomal acid, allowing its entry into the oocyte.

THE FIRST CELL OF LIFE

The meeting of the genetic heritage of the two cells forms the first cell of life. This dynamic process gives rise to the first two **blastomeres**, which are the first germ-mother cells.

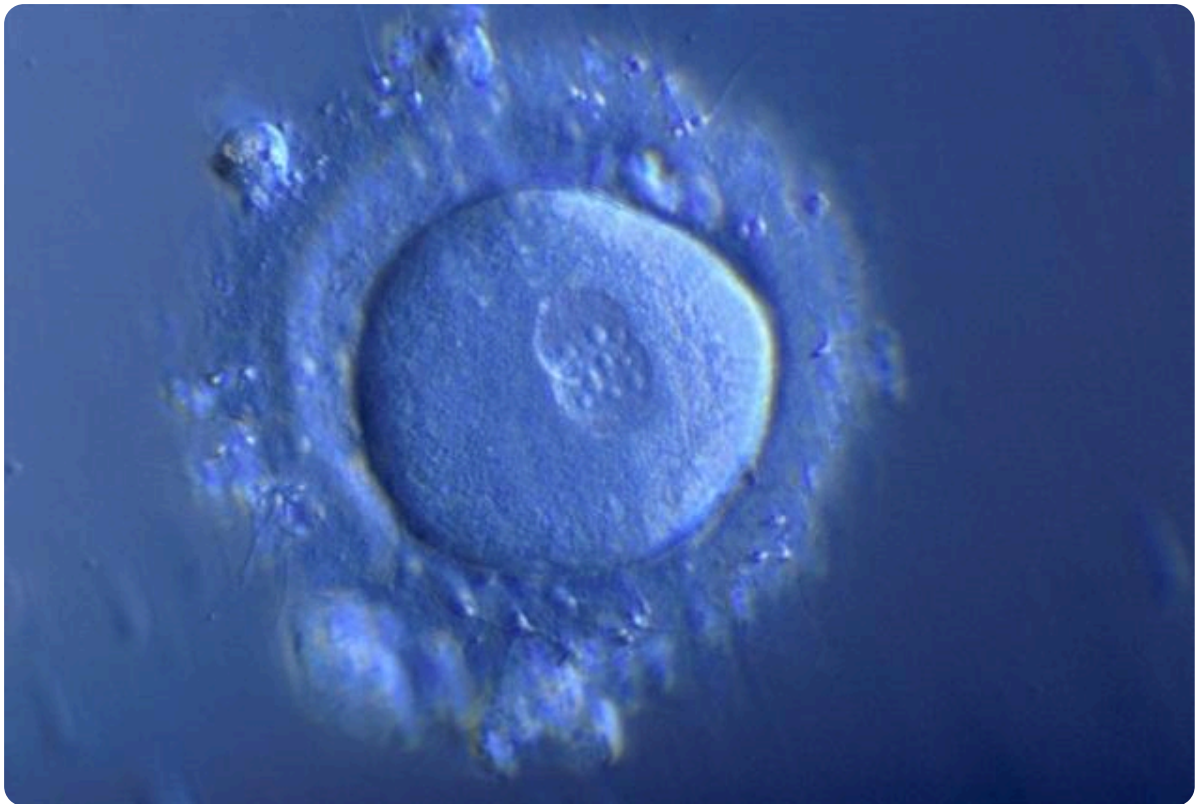


FIG. — *Fertilized egg*

This research on polarity and the first phases of embryonic development is essential for understanding the foundations of life.